# Sahil Singh

B.Tech, Mechanical Engineering, IIT Kanpur
CPI: 7.93/10 (2020 - 2024)
sahil85.10s@gmail.com | +91-9874082672
GitHub: Sahilsingh0808 | LinkedIn: sahilsingh0808

## Work Experience

**Bolto (YC S23)** | Software Engineer | Remote May'26 - Present

- Built the employee termination system end to end on a payroll and HR platform for US companies: scheduled terminations applied by daily background jobs, final payouts, transfer of duties, permission checks and a full audit trail.
- Shipped payroll groups and pay schedules: moving workers between groups, a second recurring payday, reopening payroll runs and changing pay dates.
- Built timesheet-to-payroll draft sync, offer-letter flows, and onboarding gates for companies migrating payroll in (payroll-provider enrollment, filing authorisation). Next.js, TypeScript, tRPC, Drizzle, Postgres, Trigger.dev, Redis.

**Spine AI (YC S23)** | SDE | Remote Aug'25 - Apr'26

- Architected the backend orchestration engine powering Spine Swarm, coordinating 1 to 10 specialized AI agents per task with 500+ agents running concurrently for parallel multi-step workflows at production scale.
- Built intelligent model routing dynamically selecting across 300+ LLMs per execution step, processing 106,000+ intelligence nodes cumulatively to reduce single-model bias and improve output reliability across reasoning, coding and retrieval tasks.
- Designed backend APIs powering 6,500+ active canvas environments with real-time collaborative agent execution, artifact persistence, branching, and full inspectability of intermediate agent state across concurrent user sessions.
- Went 0 to 1 across the full stack (React canvas, backend, DevOps), scaling Spine to 10k+ users and 151 paying customers (2.5% conversion) with zero-downtime blue-green deployments across AWS, GCP and Azure.

**SigIQ.ai** | SDE, Founding Member | Gurgaon Jun'24 - Jul'25

- Migrated backend services from PM2 to a containerized ecosystem, improving scalability and deployment consistency.
- Built a high-concurrency WebSocket microservice handling 10k concurrent requests and a Redis-powered real-time leaderboard delivering low-latency user-engagement insights across the platform's service usage rankings.
- Integrated Stripe for domestic and international payments (GRE product); architected Datadog APM with centralized logging across AI webservers, improving real-time monitoring and reducing outage response times via proactive alerting.
- Executed blue-green deployments with Docker and PM2 ensuring zero downtime; led and mentored a team of interns across backend and DevOps, delegating tasks, reviewing code, and driving continuous team development.

**SigIQ.ai** | SDE Intern | Remote | PPO Offered Oct'23 - May'24

- Built backend APIs and WebSockets in Django for AI chat interactions and a Redis leaderboard with logarithmic-time access.
- Developed a custom notification panel with FCM and Celery (bulk, CSV upload, analytics, internal testing); implemented deeplink-based article sharing via Firebase with auto-generated images stored in AWS S3.

**GIVA Indiejewels** | SDE Intern | Bangalore | PPO Offered May'23 - Jul'23

- Full-stack work on internal POS tools (Next.js, Node.js); shipped mission-critical UI flows to the React Native mobile app.
- Integrated Shopify webhooks for real-time order sync and Razorpay for secure domestic payments; discovered and patched an indefinite-OTP vulnerability via timeout invalidation to prevent unauthorized access.

## Key Projects

**AI-based Labor Room Smart Assistant** | Google AI4SG '21-'22 | Prof. Vipul Arora, EE Jan'22 - Jan'23

- Secured USD 30,000 funding from Google's AI for Social Good program (out of 1000+ global applicants); built a Flutter and Flask app with ASR for timely data documentation in low-resource labor rooms in Bihar.

**Feed Harmony** | GDSC IITK, Google Solution Challenge | Overall Manager Feb'24

- Led Product/Design, Development and ML teams to ship a mobile app tackling global hunger with predictive ML models within a 3-week sprint; contributed core mobile components and drove cross-team integration end to end.

**kronos-india** | Event-driven backtester for Indian markets | Python 2026

- Market data, signals, orders and fills flow through a single event queue, so a strategy runs the same way in a backtest as it would live on NSE.

## Awards & Leadership

- Founder & Lead, Google Developer Student Club, IIT Kanpur: grew to 1,000+ members with 5,000+ event participants; led a 9-member core team and 50 volunteers across Ideathons, Cloud Jams and Generative AI workshops.
- Bronze medal, Inter IIT Tech Meet 10.0: built ISRO's solar burst identifier tool for X-ray light curves.
- Google Code Jam 2021: Global Rank 794; Google Kick Start 2021: best rank 1544 (56k+ global participants).

## Technical Skills

**Languages:** C++, C, Python, Java, JavaScript, TypeScript, Dart, C#, SQL
**Backend & Data:** Django, Flask, FastAPI, Node.js, Express, tRPC, REST, WebSockets, Redis, MongoDB, PostgreSQL, Drizzle, Firestore, Celery, Trigger.dev
**Frontend & Mobile:** React, Next.js, React Native, Flutter, Redux, Tailwind, HTML/CSS, D3.js, Chart.js
**AI/ML:** RAG, Vector DBs, LLM Orchestration, OpenAI/Gemini/Anthropic APIs, Pandas, NumPy, Scikit-learn, ASR
**DevOps & Integrations:** AWS, GCP, Azure, Docker, CI/CD, Datadog, Firebase, Stripe, Razorpay, Shopify, FCM, Git, Linux